

ANN9: Logistic Regression using TensorFlow – Detailed Code & Viva Explanation

Cell 1: Install & Import

Code:

```
try:
    import tensorflow as tf
except ImportError:
    !pip install tensorflow
    import tensorflow as tf
```

Explanation (what to say in viva):

Here I check whether TensorFlow is installed.

If not, I install it using pip.

This ensures the code runs on any system.

This is useful in exam environments.

Cell 2: Import Libraries

Code:

```
import numpy as np
import matplotlib.pyplot as plt
```

Explanation (what to say in viva):

Numpy is used for numerical operations.

Matplotlib is used for plotting graphs.

Graphs help visualize model performance.

Cell 3: Load Dataset

Code:

```
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
```

Explanation (what to say in viva):

MNIST dataset is loaded.

It contains handwritten digit images.

Used for training and testing the model.

Cell 4: Preprocessing

Code:

```
x_train = x_train.reshape(-1, 784)/255.0
x_test = x_test.reshape(-1, 784)/255.0
```

Explanation (what to say in viva):

Images are reshaped into 1D vectors.

Values are normalized between 0 and 1.

This improves training speed and accuracy.

Cell 5: Model

Code:

```
model = tf.keras.Sequential([
    tf.keras.layers.Dense(10, activation='softmax')
])
```

Explanation (what to say in viva):

This defines logistic regression model.

Dense layer connects all inputs to outputs.

Softmax is used for multi-class classification.

Cell 6: Compile

Code:

```
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
```

Explanation (what to say in viva):

Optimizer controls learning process.

Loss function measures error.

Accuracy measures performance.

Cell 7: Train

Code:

```
history = model.fit(x_train, y_train, epochs=10)
```

Explanation (what to say in viva):

Model is trained for 10 epochs.

Each epoch means one pass through dataset.

Training improves accuracy.

Cell 8: Evaluate

Code:

```
model.evaluate(x_test, y_test)
```

Explanation (what to say in viva):

Model is tested on unseen data.

This gives test accuracy.

It checks generalization ability.

Cell 9: Plot Loss

Code:

```
plt.plot(history.history['loss'])
```

Explanation (what to say in viva):

Loss graph shows error reduction.

Decreasing curve indicates learning.

Cell 10: Plot Accuracy

Code:

```
plt.plot(history.history['accuracy'])
```

Explanation (what to say in viva):

Accuracy graph shows improvement.

Increasing curve indicates better performance.